SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)
(Approved by AICTE, New Delhi, Affiliated to JNTUK, Kakinada)
Accredited by NAAC with "A+" Grade.
Recognised as Scientific and Industrial Research Organisation
SRKR MARG, CHINA AMIRAM, BHIMAVARAM – 534204 W.G.Dt., A.P., INDIA
Estd. 1980

Regulation:R26	**I/IV-B.Tech. I -Semester**

ELECTRONICS & COMMUNICATION ENGINEERING

COURSE STRUCTURE

(With effect from 2026-27 admitted Batch onwards)

Course Code	Course Name	Category	L	T	P	Cr	C.I.E.	S.E.E.	Total Marks
A2610001	English for Technical Communication	**HS**	3	0	0	3	40	60	100
A2610002	Linear Algebra and Calculus	**BS**	3	0	0	3	40	60	100
A2610401	Applied Physics	**BS**	3	0	0	3	40	60	100
A2610005	Computational Thinking and Problem Solving using C	**ES**	3	0	0	3	40	60	100
A2610402	Elements of Electrical Engineering	**ES**	3	0	0	3	40	60	100
A2610008	Design Thinking and Innovation	**ES**	0	1	3	2.5	50	50	100
A2610403	Applied Physics Laboratory	**BS**	0	0	3	1.5	50	50	100
A2610013	Computational Thinking and Problem Solving using C Lab	**ES**	0	0	3	1.5	50	50	100
A2610404	Elements of Electrical Engineering Laboratory	**ES**	0	0	3	1.5	50	50	100
A2610018	NSS, Sports & Yoga	**MC**	0	0	2	0	50	-	50
		TOTAL	**15**	**1**	**14**	**22**	**450**	**500**	**950**

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
A2610001	HS	3	--	--	3	40	60	3 hrs

ENGLISH FOR TECHNICAL COMMUNICATION

(Common to all Branches)

Course Objectives:

1.	Enable students identify the context and thematic information from social, academic, and workplace dialogues and texts.
2.	Develop students' ability to use appropriate grammatical structures and vocabulary.
3.	Equip students with contextual vocabulary for writing.
4.	Familiarize students with reading strategies to comprehend and interpret texts.
5.	Develop students' ability to demonstrate appropriate social media etiquette.

Course Outcomes

Sl. No	Outcome	Knowledge Level
1.	Identify the thematic information from transactional dialogues and texts to construct coherent written and spoken discourses.	K2
2.	Apply appropriate grammatical structures and vocabulary to summarize texts on the related themes.	K3
3.	Apply contextual vocabulary to paraphrase texts and write letters.	K3
4.	Develop reading strategies to comprehend and interpret various texts for a global understanding.	K3
5.	Construct book/film reviews and demonstrate appropriate social media etiquette in writing, employing sentence synthesis techniques.	K3

SYLLABUS

UNIT-I (10Hrs)	**Effective Communication: Exploring Channels and Methods** **Vocabulary:** Roots, Prefixes, and Suffixes; Homonyms **Grammar:** Articles, Prepositions, Tenses. **Listening:** Listening for main ideas and supporting details, Note-taking **Speaking:** Introducing self and others, Describing people and objects, JAM **Reading:** Definition and Importance, Skills and Sub-Skills of Reading **Writing:** Cohesion and Coherence, Paragraph Writing
UNIT-II (10 Hrs)	**Eco-Consciousness: Negotiating for a Sustainable Future** **Vocabulary:** Contractions, Phrasal Verbs, Idioms & Phrases **Grammar:** Subject-Verb Agreement, Conditional Sentences **Listening:** Listening to various texts and comprehension **Speaking:** Role-plays, Extempore

	Reading: Reading for Inference, Point of View, Tone and Perspective **Writing:** Note-making, Summarizing, Paraphrasing (at basic level)
UNIT-III (10 Hrs)	**Disruptive Technologies: Ethical Implications and Future Prospects** **Vocabulary:** One-Word Substitutes, Commonly Confused Words **Grammar:** Degrees of Comparison, Active and Passive Voice, Question Tags **Listening:** Listening to stories, news, documentaries, and discussions **Speaking:** Debate, Group Discussions **Reading:** Reading the extracts/articles/ essays of the newspapers, journals, and literary texts. **Writing:** Paraphrasing (at advanced level), Letter Writing
UNIT-IV (10 Hrs)	**Entrepreneurship and Start-Up Culture: Emerging Trends** **Vocabulary:** Business/ Profession/ Career/ Job-related Vocabulary **Grammar:** Sentence Synthesis, Simple, Compound and Complex Sentences **Listening:** Listening to debates and discussions **Speaking:** PowerPoint Presentations by Individuals and Teams **Reading:** Reading general texts for global understanding **Writing:** Email Writing, Data Interpretation, Report Writing
UNIT-V (10 Hrs)	**Ethical Leadership: Integrating Social Responsibility and Core Values** **Vocabulary:** Advanced Vocabulary for GRE/TOEFL/ IELTS/CAT **Grammar:** Sentence Synthesis, Para Jumbles **Listening:** Listening to talks/speeches by eminent personalities **Speaking:** Team Discussions and Presentations **Reading:** Reading specific texts for global comprehension **Writing:** Book/Film Reviews, Social Media Etiquette
Textbook:	
1.	Chekuri, T., & Perumalla, R. K. (2026). NextGen English for engineers: A course book for technical English communication. Cengage.
Reference Books:	
1.	Technical Communication (2nd Ed.) by M. Raman & S. Sharma. Oxford University Press, 2012.
2.	Effective Technical Communication by M.A. Rizvi, Tata McGraw-Hill, 2005.
3.	English for Engineers and Technologists, Orient Blackswan 2010 (Combined edition, Vol.1 &2).
4.	The Oxford Guide to Writing and Speaking by John Seely, Oxford University Press, 2004.
e-Resources	
1.	eGyanKosh. (n.d.). National digital repository. https://egyankosh.ac.in/
2.	Read Theory. (n.d). Adaptive reading comprehension practice. https://readtheory.org/

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
A2610002	BS	3	0	--	3	40	60	3 Hrs.

LINEAR ALGEBRA & CALCULUS

(Common to All Branches of Engineering)

Course Objectives: Students are expected to learn

1.	Concepts of linear algebra and methods of solution of linear simultaneous algebraicequations.
2.	Eigen values, Eigen vectors and quadratic forms.
3.	Problems and applications of Mean value theorems
4.	Application of partial differentiation for determining maxima/ minima of functions.
5.	Concepts of double, triple integrals and its applications.

Course Outcomes: After completion of the course, the student will be able to

S.No	Outcome	Knowledge Level
1.	**Solve** a system of linear equations by using direct and indirect methods.	K3
2.	**Determine** Eigen values and Eigen vectors, execute matrix Diagonalization and reduce Q.F.'s to canonical forms.	K3
3.	**Apply** mean value theorems to analyze function behaviour and solve real life problems.	K3
4.	**Apply** the concepts of partial differentiation in various engineering applications.	K3
5.	**Apply** the concepts of multiple integrals to calculate areas, volumes of geometrical regions and in engineering applications.	K3

SYLLABUS

UNIT-I (10Hrs)	**Linear System of Equations:** Rank of a matrix by echelon form, normal form. Inverse of a matrix by Gauss-Jordan method. Consistency of linear equations: Homogeneous and Non-Homogeneous systems. Gauss elimination method, Jacobi and Gauss Seidel Iteration Methods.
UNIT-II (10 Hrs)	**Eigen Values, Eigen Vectors and Quadratic Forms:** Eigen values and Eigen vectors of a matrix, their properties, Cayley-Hamilton Theorem (without proof), Inverse and powers of a matrix by Cayley – Hamilton theorem, Diagonalization of a matrix, Quadratic forms and Nature of the Quadratic Forms, Reduction of a Quadratic form to canonical form by Orthogonal Transformation and Singular Value Decomposition.
UNIT-III (10 Hrs)	**Single Variable Calculus:** Mean Value Theorems: Rolle's Theorem, Lagrange's mean value theorem, Geometrical interpretation of these theorems, Cauchy's mean value theorem, Taylor's and Maclaurin's

	theorems (without proof) with Lagrange's form of remainder, Taylor's and Maclaurin series.
UNIT-IV (10 Hrs)	**Partial differentiation and Applications:** Functions of several variables, Continuity and Differentiability, Partial derivatives, Euler's theorem, Total derivative, chain rule, Taylor's and Maclaurin's series expansion of function of two variables, Jacobians, Functional dependence, Maxima and Minima of function of two variables without constraints and with constraints - Method of Lagrange multipliers.
UNIT-V (10 Hrs)	**Multiple Integrals:** Double integrals, change of order of integration, triple integrals, change of variables from Cartesian to polar, cylindrical and spherical coordinates. Finding areas (by double integrals) and volumes (by double integrals and triple integrals).

Textbooks:	
1.	HigherEngineeringMathematics,B.S.Grewal,KhannaPublishers,2017, 44th Edition
2.	AdvancedEngineeringMathematics,ErwinKreyszig,JohnWiley&Sons,2018,10thEdition.
Reference Books:	
1.	ThomasCalculus,GeorgeB.Thomas,MauriceD.WeirandJoelHass,PearsonPublishers,2018, 14thEdition.
2.	Higher Engineering Mathematics, B.V.Ramana, McGraw HillEducation,2017.
3.	Advanced Modern Engineering Mathematics, Glyn James, Pearson publishers, 2018, 5th Edition.
4.	Advanced Engineering Mathematics, Micheael Greenberg, Pearson publishers, 9th edition.
5.	Higher Engineering Mathematics, H. K Das, Er. Rajnish Verma, S. Chand Publications,2014, Third Edition (Reprint 2021).
6.	Advanced Engineering Mathematics by H. K Dass, S. ChandPublications,2022, Twenty Two EditionReprint 2022).
7.	Numerical Python: A Practical Techniques Approach for Industry, Robert Johansson, APress, 2015.
e-Resources	
1.	https://en.wikipedia.org/wiki/System_of_linear_equations
2.	https://en.wikipedia.org/wiki/Eigenvalues_and_eigenvectors
3.	https://www.math.hmc.edu/calculus/tutorials/eigenstuff/
4.	https://en.wikipedia.org/wiki/Quadratic_form
5.	https://en.wikipedia.org/wiki/Calculus
6.	https://en.wikipedia.org/wiki/Partial_derivative
7.	https://www.whitman.edu/mathematics/calculus_online/section14.03.html
8.	https://en.wikipedia.org/wiki/Multiple_integral
9.	http://tutorial.math.lamar.edu/Classes/CalcIII/MultipleIntegralsIntro.aspx

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
A2610401	BS	3	--	--	3	40	60	3 Hrs.

APPLIED PHYSICS

(For ECE)

Course Objectives:

1.	To provide knowledge on wave optics concepts such as interference, diffraction, and polarization with relevant engineering applications.
2.	To understand the basic concepts of Laser light sources and working principle of optical fibers and identify the engineering applications of lasers and optical fibers in communication.
3.	To familiarize students with electrical properties of metals and semiconductors and relevant applications in electronic devices.
4.	To equip students the fundamentals of quantum mechanics and principles relevant to engineering applications such as quantum computing.
5.	To impart knowledge on electromagnetic field theory and its applications.

Course Outcomes: After successful completion of this course, the student will be able to:

S.No.	Outcome	Knowledge Level
1.	Explain the need of coherent sources and apply the principles of Interference, diffraction and polarization of light in engineering applications.	K3
2.	Analyze the working of lasers, fiber optics, and advanced sensing systems in optical communication and signal processing techniques.	K4
3.	Analyze the electrical properties of metals and classify the energy bands of semiconductors and identify the type of semiconductor using Hall effect. Identify the applications of semiconductors in electronic devices.	K4
4.	Apply the concepts of quantum mechanics and quantum computing along for secure communication applications.	K3
5.	Apply Maxwell's equations and electromagnetic field concepts for antenna-based systems.	K3

SYLLABUS

UNIT-I (10Hrs)	**APPLIED WAVE OPTICS** **Interference:** Superposition Principle – Coherence – Types of Interference – Thin-film interference (reflection geometry) – Newton's rings – Determination of wavelength and refractive index. **Diffraction and Optical Surveying:** Fresnel and Fraunhofer diffraction – Fraunhofer's diffraction due to single slit (quantitative), double slit & N-slits (qualitative) – Diffraction Grating and grating spectrum – Dispersive and Resolving power of a grating (qualitative). **Polarization:** Types of polarization – Nicol's prism – Half-wave and Quarter-wave plates.

UNIT-II (9 Hrs)	**QUANTUM PHOTONICS & ADVANCED SENSING** **Lasers:** Principles of Absorption – Spontaneous and Stimulated Emission – Population Inversion – Three level and Four level laser systems – Construction and working of Ruby and He-Ne Laser – Applications (qualitative). **Fiber Optics:** Propagation mechanism – Numerical Aperture – Acceptance angle – Types of optical fibres – Classifications and Modes of optical fibres.
UNIT-III (10 Hrs)	**ELECTRICAL PROPERTIES OF METALS & SEMICONDUCTORS** **Electrical Properties of Metals:** Classical free electron theory – merits and demerits, Quantum free electron theory – equation for electrical conductivity – Fermi-Dirac statistics – Fermi level and variation with temperature. **Semiconductors:** Formation of energy bands – Classification of solids - Intrinsic semiconductors – Density of charge carriers – Electrical conductivity – Fermi level – extrinsic semiconductors: density of charge carriers (qualitative) – dependence of Fermi energy on carrier concentration and temperature – Drift and diffusion currents – Einstein's equation – Hall effect – Hall coefficient – Applications of Hall effect.
UNIT-IV (10 Hrs)	**INTRODUCTION TO QUANTUM MECHANICS & QUANTUM COMPUTING** **Quantum Mechanics:** de Broglie concept of matter waves – Heisenberg's uncertainty principle – Wave function – Postulates of quantum mechanics – Schrödinger's time dependent and time-independent wave equations -–Particle in a one-dimensional potential well – Quantum tunneling (basic idea). **Quantum Computing:** Introduction to quantum computing – Classical bits vs qubits – Superposition and entanglement – Basic quantum operations – Hadamard gate – Pauli-X gate – Idea of quantum measurement.
UNIT-V (9 Hrs)	**INTRODUCTION TO ELECTROMAGNETIC THEORY** Introduction – Wave Equation in differential form in free space – Vector Analysis – Divergence and Curl of electric field and magnetic field (static) – Gauss' divergence theorem – Stokes' theorem – Derivations of the four Maxwell Equations and their significance. Maxwell's equations in vacuum, Energy transport and Poynting Vector.

Textbooks:	
1.	Avadhanulu, M.N., Kshirsagar, P.G., & Murthy, T.V.S.A., A textbook of Engineering Physics, S. Chand Publishing (2019)
2.	P.K. PalaniSamy, Engineering Physics, SciTech Publications (2019)
Reference Books:	
1.	D. K. Bhattacharya & Poonam Tandon, Engineering Physics, Oxford Press (2015)
2.	S. O. Pillai, Solid State Physics, New Age International (2005)
3.	D. J. Griffiths, Introduction to Electrodynamics, Cambridge (2023)
4.	Nielsen & Chuang, Quantum competition & Quantum information, Cambridge (2010 Edition)

e-Resources	
1.	NPTEL – nptel.ac.in
2.	MIT OpenCourseWare – ocw.mit.edu
3.	HyperPhysics – hyperphysics.phy-astr.gsu.edu
4.	Feynman Lectures – feynmanlectures.caltech.edu
5.	PhET Simulations – phet.colorado.edu

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
A2610005	ES	3	--	--	3	40	60	3 Hrs

COMPUTATIONAL THINKING AND PROBLEM SOLVING USING C

(Common to AIML, CIC, CSD, CSE, CSIT, ECE, IT)

Pre-requisite: Mathematics and logical reasoning at 10+2 level. No prior programming experience is assumed.

Course Objectives: The main objectives of the course are to

1.	Familiarize students with computers, computational thinking, and the process of algorithmic problem solving.
2.	Build strong programming fundamentals using the C language, including operators and control structures.
3.	Develop the ability to design and implement solutions using arrays, strings, and user- defined data types.
4.	Impart working knowledge of pointers, recursion, dynamic memory allocation, functions, and file handling for building modular, reusable programs.

Course Outcomes: *At the end of the course, student will be able to*

S.No	Outcome	Knowledge Level
1.	Apply computational thinking strategies to construct algorithms, flowcharts, and implement them as C programs.	K3
2.	Apply selection and iterative control structures along with arrays to solve programming problems.	K3
3.	Analyze problems involving strings and user-defined datatypes to construct suitable C solutions.	K4
4.	Apply pointers, dynamic memory allocation, and recursive techniques to design modular C programs using functions.	K3
5.	Apply preprocessor directives and file handling techniques to perform file input-output operations in C programs.	K3

SYLLABUS

UNIT-I (10Hrs)	**Computational Thinking and Introduction to C** **Computational Thinking:** Decomposition, Pattern Recognition, Abstraction, Algorithm Design; representing solutions as algorithms, pseudocode, and flowcharts; Number Systems and Data Representation (integers, reals); Overview of Computer Organization and Program Execution. **Introduction to C:** Background, Structure of a C Program, Identifiers, Keywords, Variables, Constants, Basic Data Types, Input/Output; Expressions, Operators, type of operators- arithmetic, logical, relational, assignment, conditional, increment,

	decrement, etc., unary, binary, ternary, Operator Precedence and Associativity, Type Conversion; Scope, Storage Classes, and Type Qualifiers. **Bitwise Operators:** Logical Bitwise Operators, Shift Operators.
UNIT-II (10 Hrs)	**Selection and Decision-Making**: Two-Way and Multi-Way Selection (if-else, switch), Nested Conditionals. **Repetition:** Pre-test and Post-test Loops, Counter- and Event-Controlled Loops, for/while/do-while, break and continue, Nested Loops; Programming Applications combining Selection and Repetition. **Arrays:** One-Dimensional and Two-Dimensional Arrays, Multidimensional Arrays, Array-based Problem Solving (searching, sorting, averaging).
UNIT-III (10 Hrs)	**Strings:** String Representation, Standard String-Handling Functions, Arrays of Strings, String-to-Data Conversion. **User-Defined Types:** typedef, Enumerated Types, Structures, Unions, and Nested Structures; Programming Applications using Structures and Arrays of Structures
UNIT-IV (10 Hrs)	**Pointers:** Pointer Fundamentals, Pointer-to-Pointer, Pointer Arithmetic, Pointers and Arrays, Array of Pointers, Compatibility and l-value/r-value concepts. Dynamic Memory Allocation: malloc, calloc, realloc, free. **Functions:** Designing Structured/Modular Programs, Function Declaration and Definition, Parameter Passing (by value, by reference using pointers), Passing Arrays and Pointers to Functions, Standard Library Functions, Inter-Function Communication. **Recursion:** Recursive vs. Iterative Solutions, Designing Recursive Algorithms, Recursion with Arrays and Pointers, Efficiency Considerations.
UNIT-V (10 Hrs)	**Preprocessor**: Directives and Macros. **File I/O:** Streams, Text vs. Binary Files, Standard Library File Functions, formatted and Character I/O, Sequential File Processing.
Textbooks:	
1.	Behrouz A. Forouzan & Richard F. Gilberg, Computer Science: A Structured Programming Approach Using C, 3rd Edition, Cengage Learning, ISBN-13: 978-0-534-49132-1.
2.	Brian W. Kernighan & Dennis M. Ritchie, The C Programming Language, 2nd Edition, Pearson Education / Prentice Hall India, 1988.
Reference Books:	
1.	Sumitabha Das, "Computer Fundamentals and Programming", 18th Edition, McGraw Hill Education, 2018
2.	Ashok N. Kamthane, Amit Ashok Kamthane, "Programming in C", 4th Edition, Pearson Education, 2025. ISBN: 978-9332543633.
3.	Pradip Dey, Manas Ghosh, "Computer Fundamentals and Programming in C", 2nd Edition, Oxford University Press, 2013. ISBN: 978-0198068842

e-Resources	
1.	NPTEL – Programming in C, IIT Kanpur: https://onlinecourses.nptel.ac.in/noc21_cs81/preview
2.	Geeks for Geeks – C Programming Language: https://www.geeksforgeeks.org/c-programming-language/
3.	Programiz – Learn C Programming: https://www.programiz.com/c-programming
4.	cplusplus.com – C Standard Library Reference: https://cplusplus.com/reference/clibrary/
5.	TutorialsPoint – C Programming: https://www.tutorialspoint.com/cprogramming/index.html
6.	W3Schools – C Tutorial: https://www.w3schools.com/c/

Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
A2610402	**ES**	**3**	**--**	**--**	**3**	**40**	**60**	**3 Hrs.**

ELEMENTS OF ELECTRICAL ENGINEERING

(For ECE)

Course Objectives: Students will learn about

1.	The various aspects of basic electrical quantities, generators, alternators, and power generation sources.
2.	The various aspects of transmission and distribution of electrical energy.
3.	The utilization of electricity in various electrical motors and their principles of operation and applications.
4.	The power conversion and energy storage of electricity
5.	The electrical hazards, electrical safety measures and equipment protection devices

Course Outcomes: At the end of the course, the students will be able to

S.No	Outcome	Knowledge Level
1.	Apply basic electrical concepts and the principles of conventional and non-conventional power generation.	K3
2.	Identity different components of transmission and distribution substations.	K3
3.	Apply energy conversion principles to understand operation of electrical machines	K3
4.	Apply basic knowledge to understand operation of rectifier, Inverter, batteries and uninterrupted power supply	K3
5.	Understand and apply the Electrical safety measures while handling electrical equipment.	K3

SYLLABUS

UNIT-I (10Hrs)	**ELECTRICITY GENERATION** Basic electrical quantities -Voltage, Current, Power and energy, DC Generator - principle of operation, 3 – Phase Alternator - principle of operation, frequency and rms value of sinusoidal voltage. Major sources of electricity generation: Schematics of conventional power plants (Thermal and Hydro), Non-conventional sources (solar and wind) [Elementary Treatment only].
UNIT-II (10 Hrs)	**TRANSMISSION AND DISTRIBUTION OF ELECTRICITY** Transmission of Electrical Energy: Layout of power system, Overhead lines and cables, Power transmission at high voltage, Transformer - Working principle, Construction, Distribution of electrical energy - Schematics diagrams of radial and ring main distribution, Substations - Layout, equipment and their purpose.
UNIT-III (10 Hrs)	**ELECTRICAL ENERGY CONVERSION** Conversion to mechanical energy -Classification of Electrical motors and their applications, DC motor - Working principle, Torque equation, servo motor, AC motor - Working principle of 3-phase Induction motor, slip. Construction and Principle of operation of Stepper motors and BLDC motors.

UNIT-IV (10 Hrs)	**POWER CONVERTERS AND STORAGE** Need of power conversion, Rectifier - Single phase full wave diode bridge rectifier with C-filter, rectifier applications, Inverter- Single phase full bridge inverter operation with R load, Inverter Applications, Electricity storage - Batteries, types of batteries, Lead acid battery, Li ion batteries, Ratings and basic parameters (SOC, DOD, SOH and C-rate), battery pack, schematic diagram of Uninterrupted power supply (UPS).
UNIT-V (10 Hrs)	**ELECTRICAL SAFETY AND EQUIPMENT PROTECTION** Hazards in electrical systems, Different types of hazards, Electric Shock, Electrical safety measures, Earthing – Pipe & Plate earthing. Domestic Protective Devices - Fuses and their ratings, Miniature Circuit Breaker (MCB), Earth Leakage Circuit Breaker(ELCB), Power ratings of different domestic loads - Fans, Lights, Air conditioners, Refrigerators, etc.
Textbooks:	
1.	D.P. Kothari & I.J. Nagrath, Basic Electrical Engineering, 4th Edition, McGraw Hill Education, 2019.
2.	V. K. Mehta and Rohit Mehta, Principles of Electrical Engineering and Electronics, 2nd Edition, S. Chand & Company Ltd., New Delhi. 2015.
Reference Books:	
1.	B. L. Theraja, Fundamentals of Electrical Engineering and Electronics (LPSPE), 1st edition of the LPSPE version, S. Chand & Company Ltd., New Delhi. 2022.
2.	V. Del Toro, Electrical Engineering Fundamentals, 2nd ed., Englewood Cliffs, NJ: Prentice-Hall, 1986.
3.	S. K. Sahdev, Basic Electrical Engineering, 1st Edition, Pearson India Education Services Pvt. Ltd., New Delhi, 2015, ISBN: 978-93-325-4216-7.
4.	C. L. Wadhwa, Generation, Distribution and Utilization of Electrical Energy, 3rd Edition, New Age International Pvt. Ltd., New Delhi, 2015.
5.	Iqbal Husain, Electric and Hybrid Vehicles: Design Fundamentals, 2nd Edition, CRC Press, Taylor & Francis Group, Boca Raton, 2011, ISBN: 978-1-4398-1175-7.
6.	D. C. Kulshreshtha, Basic Electrical Engineering, 1st Edition, Tata McGraw Hill, New Delhi, 2009, ISBN: 978-0-07-106864-2.
7.	U.A. Bakshi and V.U. Bakshi, Basic Electrical Engineering, 2nd Revised Edition, Technical Publications, Pune, 2009, ISBN: 978-81-8431-694-0.
e-Resources	
1.	https://nptel.ac.in/courses/108102185
2.	https://nptel.ac.in/courses/103103206

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
A2610008	**ES**	**--**	**1**	**3**	**2.5**	**50**	**50**	**3 Hrs.**

DESIGN THINKING AND INNOVATION

(Common to AIML, CIC, CIVIL, CSE, CSBS, ECE, EEE, MECH)

Course Objectives: At the end of the course, the teacher should be able to:

1	Introduce students to the principles, mindset and methodologies of **Design Thinking and human-centered problem solving**.
2	Develop students' ability to **observe, empathize, analyze user needs and formulate well-defined engineering and societal problems**.
3	Develop students' **creative and innovative thinking abilities** through systematic ideation, concept generation and evaluation techniques.
4	Enable students to **develop, prototype, test and iteratively improve solutions** using user feedback and appropriate physical/digital tools.
5	Develop students' ability to identify innovation opportunities, create value propositions, understand basic IPR and communicate innovative solutions effectively.

Course Outcomes: *After successful completion of the course, student will be able to*

S. No	Outcome	Knowledge Level
1	Explain the principles, processes and mindset of Design Thinking for engineering problem-solving and innovation	K2
2	Apply empathy, observation and problem-framing techniques to identify user needs and define engineering problems	K3
3	Apply creative and systematic ideation techniques to generate and select innovative solution concepts	K3
4	Analyze solution concepts and prototypes using user feedback and appropriate design criteria for iterative improvement	K4
5	Analyze the innovation potential of solutions considering value proposition, feasibility, sustainability, societal impact and basic IPR aspects	K4

1 **UNIT I — Design Thinking Foundations & Empathy**

Introduction to Design Thinking ; Design thinking mindset and principles ; Design thinking vs. conventional engineering problem-solving ; Human-centered design ; Five-stage design thinking process ; Creativity, curiosity and observation ; Understanding users and stakeholders ; Empathy and user research ; Interviews, observation and contextual inquiry ; Empathy mapping ; Personas and user journeys ; Identifying explicit and latent/unarticulated needs ; Ethical considerations in user research

Practical Activities

- Observation of a real-life problem
- Conducting user interviews

	• Preparing an empathy map • Developing a persona • Preparing a user journey map **Unit Outcome** Students will be able to **understand users, identify their needs and translate observations into meaningful design opportunities**
2	**UNIT II — Problem Definition, Analysis & Opportunity Identification** From observations to insights ; Data synthesis and affinity mapping ; Identifying patterns and pain points ; Problem statements ; Point-of-View (POV) statements ; "How Might We" (HMW) questions ; Stakeholder mapping ; 5 Whys and root-cause analysis ; Problem prioritization ; Need–Opportunity mapping ; Problem validation ; Design constraints and specifications ; Social, environmental and sustainability considerations ; Identifying opportunities for innovation **Practical Activities** • Affinity mapping of interview data • Develop three alternative problem statements • Prepare POV and HMW statements • Stakeholder map • Root-cause analysis • Select and validate one problem/opportunity **Unit Outcome** Students will be able to **analyze user research, identify root causes, frame well-defined problems and recognize opportunities for innovative solutions**.
3	**UNIT III — Creativity, Ideation & Concept Development** Creativity and innovation mindset; Divergent and convergent thinking ; Brainstorming and brainwriting ; SCAMPER ; Mind mapping ; Reverse brainstorming ; Analogical thinking ; Biomimicry ; Lateral thinking ; Forced connections ; Morphological analysis ; TRIZ — basic principles ; Individual and collaborative ideation ; AI-assisted ideation: opportunities and limitations ; Idea clustering and screening ; Impact–feasibility matrix ; Concept selection ; Design criteria and decision matrix **Practical Activities** • Generate 20–30 ideas for a selected problem • SCAMPER exercise • Brainwriting exercise • Biomimicry-based ideation • AI-assisted idea generation and critical evaluation • Prepare an idea-selection matrix • Select the top 2–3 concepts **Unit Outcome** Students will be able to **generate diverse solution concepts using creative techniques and systematically select promising ideas based on defined criteria**.

4	**UNIT IV — Prototyping, Experimentation & Validation** Concept development ; Prototype mindset ; Types of prototypes: Paper prototype, Physical prototype, Digital prototype, Storyboard, Role-play, Service prototype ; Low-fidelity and high-fidelity prototypes ; Rapid prototyping ; "Fail fast, learn fast" ; Minimum Viable Prototype (MVP) ; Design experimentation ; Prototype evaluation ; User testing ; Think-aloud method ; Feedback collection ; Usability and functionality testing; Iterative design ; Design critique ; Documentation of iterations ; Use of CAD, simulation and AI-assisted visualization tools **Practical Activities** • Develop a low-fidelity prototype • Prepare a storyboard • Build a physical/digital prototype • Conduct user testing • Collect structured feedback • Modify the prototype • Prepare an iteration log
5	**UNIT V — Innovation, Value Creation & Communication** **Innovation: meaning, characteristics and importance:** Invention vs. innovation ; Incremental, disruptive and radical innovation ; Product, process, service and social innovation ; Technology-driven vs. need-driven innovation ; Sustainable and frugal innovation ; Innovation opportunities and emerging technologies ; From prototype to innovation ; Desirability, feasibility and viability ; Value Proposition ; Value Proposition Canvas ; Introduction to Business Model Canvas ; Customer segments and stakeholder value; Innovation validation; Market/user validation ; Intellectual Property Rights — introduction; Patents, copyright, trademarks and design protection ; Responsible innovation ; Inclusive and sustainable innovation ; Innovation ecosystem and entrepreneurship ; Communicating innovative ideas ; Innovation pitch **Practical Activities** 1. Classify existing products/services as incremental, disruptive or radical innovations. 2. Identify three innovation opportunities from a campus/community problem. 3. Prepare a **Value Proposition Canvas**. 4. Validate the proposed innovation with potential users. 5. Prepare an introductory **Business Model Canvas**. 6. Identify possible IPR protection. Prepare and deliver a **3-minute Innovation Pitch**
Text Books:	
1	Chandan Adhikari, Prabir Kumar Das & Tanay Pramanik, Textbook of Design Thinking and Innovation, S. Chand Publishing, 2026.
2	Shalini Rahul Tiwari & Rohit Rajendra Swarup, Design Thinking: A Comprehensive Textbook, Wiley India, 2024
Reference Books:	
1	Soni, Pavan, Design Your Thinking: The Mindsets, Toolsets and Skill Sets for Creative Problem-Solving, Penguin Random House India.

2	Jauhari, Vinnie & Sudhanshu Bhushan, Innovation Management, Oxford University Press India.
Web References:	
1	Atal Innovation Mission, NITI Aayog – Innovation and Design Thinking Resources. https://aim.gov.in/
2	Institution's Innovation Council (IIC), Ministry of Education, Government of India, https://iic.mic.gov.in/
3	Startup India, Government of India – Innovation and Entrepreneurship Ecosystem, https://www.startupindia.gov.in/
4	National Innovation Foundation – India – Grassroots and Frugal Innovation, https://nif.org.in/
5	Stanford d.school – Design Thinking Bootleg and Design Thinking Resources, https://dschool.stanford.edu/resources/design-thinking-bootleg
6	IDEO.org – Human-Centered Design and Social Innovation Resources, https://www.ideo.org/
7	IDEO U – Design Thinking and Innovation Resources, https://www.ideou.com/pages/design-thinking
8	NPTEL – Online Courses on Design, Innovation, Creativity and Entrepreneurship, https://nptel.ac.in/
9	SWAYAM – Online Courses on Innovation, Entrepreneurship and Design Thinking, https://swayam.gov.in/

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
A2610403	BSC	--	--	3	1.5	50	50	3 Hrs.

APPLIED PHYSICS LABORATORY

(For ECE and Allied Branches of Engineering)

Course Objectives:

1	To impart hands on experience to the students entering Engineering & Technology education about handling equipment/instruments and use them in experimentation.
2	To make the students understand theoretical aspects of various phenomena experimentally.

Course Outcomes:

S.No.	Outcome	Knowledge Level
1	Apply fundamental principles of Physics to conduct engineering laboratory experiments.	K3
2	Get hands on experience in setting up experiments and using the instruments/equipment individually and conduct experiments.	K3
3	Apply graphical and analytical methods to analyze experimental data and determine physical parameters.	K3
4	Get introduced to using new/advanced technologies and understand their significance .	K3

List of Experiments

(Any **10** of the following listed experiments)

(Out of which any **TWO** experiments may be conducted in virtual mode)

1	Determination of thickness of thin object by wedge method.
2	Determination of radius of curvature of a given plano convex lens by forming Newton's rings.
3	Determination of wavelengths of different spectral lines in mercury spectrum using diffraction grating in normal incidence configuration.
4	Determination of resolving power of telescope.
5	Determination of dielectric constant of given capacitor by charging and discharging method.
6	Study the variation of B versus H by magnetizing the magnetic material (B-H curve).
7	Determination of temperature coefficients of a thermistor.
8	Determination of wavelength of Laser and grating element.
9	Estimation of Planck's constant using photoelectric effect.
10	Determination of the Resistivity of semiconductor by four probe method.
11	Determination of the energy gap of a semiconductor using p-n junction diode.
12	Study the variation of magnetic field along the axis of a current carrying circular coil by Stewart & Gee's Method
13	Determination of Hall voltage and Hall coefficient of a given semiconductor using Hall Effect.
14	Determination of Acceleration due to gravity and Radius of gyration using Compound pendulum.

15	Determination of magnetic susceptibility by Kundt's tube method
16	Verification of Brewster's law.
17	Determination of frequency of A.C. supply using Sonometer.
18	Determination of frequency of electrically maintained tuning fork by Melde's apparatus.
19	Determination of Rigidity modulus of material of a wire-dynamic method (Torsional pendulum).
Reference Books:	
1	S. Balasubramanian, M.N. Srinivasan "A Text Book of Practical Physics"- S Chand Publishers, 2017.

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
A2610013	ESC	0	0	3	1.5	50	50	3 Hrs

COMPUTATIONAL THINKING AND PROBLEM SOLVING USING C LAB

(Common to AIDS, AIML, CIC, CSBS, CSD, CSE, CSIT, ECE, IT)

Pre-requisite: - Basic understanding of computer fundamentals and problem-solving concepts

Course Objectives: The main objectives of the course are to

1	Apply the principles of C language and computational thinking techniques in problem solving.
2	Design flowcharts and algorithms, and to develop the ability to trace and debug programs.
3	Design and develop C programs using arrays, strings, pointers, recursion, and functions.
4	Apply dynamic memory allocation, file operations, and preprocessor commands in program design.

Course Outcomes:

S.No	Outcome	Knowledge Level
1	Apply computational thinking to construct algorithms and flowcharts for a given Problem and translate them into C programs.	K3
2	Develop C programs using operators, selection statements and loops to implement a specified logic.	K3
3	Develop C programs that manipulate arrays, strings, structures and pointers to solve computational problems.	K3
4	Construct modular programs using user-defined functions and recursion and perform file input-output operations.	K3
5	Analyze a C program by tracing and debugging it and evaluate alternative Implementations for correctness and efficiency.	K4

SYLLABUS

1	**Exercise 1: Getting Started with** 1. Write, compile and run a C program that prints "Hello, World!" on the screen. Identify the source file, the object file and the executable produced, and record the command used at each stage. 2. Write a C program that declares variables of type int, float, char and double, reads a value for each from the user, and displays each value together with the size of its type using size of. 3. Write a C program to read the height and width of a rectangle from the user and compute its perimeter and area.
2	**Exercise 2: Operators and Expressions** 1. Write a C program to compute the distance between two points whose coordinates are entered by the user. 2. Write a C program that reads four integers p, q, r, s and, using relational and logical operators, prints 'Correct values' if q > r, s > p and (r + s) > (p + q); otherwise 'Wrong values'. 3. Write a C program to compute simple and compound interest for values entered by the user.

3	**Exercise 3: Selection and Decision Making** 1. Write a C program to read a character and classify it as a vowel, consonant, digit or special symbol. 2. Write a C program to find the largest of three numbers using nested if-else. 3. Write a menu-driven C program to compute the area of various geometrical shapes (circle, rectangle, triangle, square) using switch-case.
4	**Exercise 4: Loops and Series** 1. Write a C program to display the first n even natural numbers and their sum. 2. Write a C program to reverse a given integer and check whether it is a palindrome. 3. Write a C program to display the sum of the harmonic series 1 + 1/2 + 1/3 + ... up to n terms.
5	**Exercise 5: Nested Loops and Number Patterns** 1. Write a C program to print the multiplication table pyramid for numbers 1 to n using nested loops. 2. Write a C program to check whether a given number is prime, then extend it to print all prime numbers and all Armstrong numbers below.
6	**Exercise 6: One-Dimensional Arrays** 1. Write a C program to print all unique elements of an array and implement linear search to find a given element in the array. 2. Write a C program to separate the odd and even elements of an array into two different arrays and implement binary search to find a given element in a sorted array. 3. Write a C program to sort the elements of an array in ascending order using selection sort and compare the efficiency of linear search vs. binary search on the sorted array by counting the number of comparisons each method takes.
7	**Exercise 7: Matrices** . Write a C program to perform addition of two square matrices and display the resultant matrices. . Write a C program to multiply two square matrices and display the resultant matrices. . Write a C program to find the transpose of a given matrix.
8	**Exercise 8: Strings:** Write C programs to perform the following without using string handling functions • **Find the length** of a given string. • **Copy** one string into another. • **Concatenate** two strings. • **Compare** two strings and display whether they are equal, or which one is greater. • **Reverse** a given string (using a separate swap function to exchange characters). • **Search for a character** in a string and display its position (or indicate if not found). • **Count the number of vowels, consonants, and words** in a given string
9	**Exercise 9: Pointers Fundamentals** 1. Write a C program to demonstrate the use of the & (address-of) and * (value-at-address) operators. 2. Write a C program to add two numbers using pointers. 3. Write a C program to swap two values using call by reference.

10	**Exercise 10: Pointers and Dynamic Memory** . Write a C program to dynamically allocate memory for an array using malloc, find the largest element and the sum of all elements in it, then repeat using calloc and note the difference between the two. . Write a C program to count the number of vowels in a string using a pointer.
11	**Exercise 11: User-Defined Functions** . Write a C program to check whether a number is prime using a user-defined function. . Write a C program to compute the GCD and LCM of two numbers using functions. . Write a C program to demonstrate a function that returns a pointer.
12	**Exercise 12: Recursion** . Write a C program to compute the factorial and the n-th Fibonacci term using recursion. . Write a C program to compute the sum of digits of a number using recursion. . Write a C program to convert a decimal number to its binary equivalent using a recursive function.
13	**Exercise 13: File Handling** . Write a C program to append multiple lines at the end of an existing text file. . Write a C program to copy the contents of one file into another file. . Write a C program to count the number of characters, words and lines in a text file.
Reference Books: **Web References:**	
1	NPTEL – Programming in C, IIT Kanpur: https://onlinecourses.nptel.ac.in/noc21_cs81/preview
2	Virtual Labs – Problem Solving Lab, IIIT Hyderabad: https://ps-iiith.vlabs.ac.in/
3	GeeksforGeeks – C Programming: https://www.geeksforgeeks.org/c-programming-language/

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
A2610404	ES	--	--	3	1.5	50	50	3 Hrs

ELEMENTS OF ELECTRICAL ENGINEERING LABORATORY

(For ECE)

Course Objectives: Student will learn about

1	The performance of DC Machines
2	The performance of AC machines.
3	The characteristics of solar panel
4	The various electrical quantity measurement in different power supply systems.
5	The safety practices in electrical systems

Course Outcomes: At the end of the course, students will be able to

S.No	Outcome	Knowledge Level
1	Analyze the performance of DC Machines.	K4
2	Analyze the performance of AC machines.	K4
3	Conduct experiment to obtain I – V Characteristics of Solar panel.	K3
4	Measure voltage, current, power and energy in different electrical power supply systems.	K3
5	Analyze the safety measures in handling electrical equipment.	K4

List of Experiments

1	Obtain the Open Circuit Characteristic (OCC) of DC shunt Generator.
2	Measurement of Solar Power Output.
3	Brake test on DC shunt motor.
4	Load test on single-phase transformer.
5	Verification of Miniature Circuit Breaker operation.
6	Measurement of Energy using single phase energy meter.
7	Speed control of D.C. shunt motor.
8	Obtain the Open Circuit Characteristics (OCC) of a 3 - Phase Alternator.
9	Load test on Three-phase Squirrel Cage Induction Motor.
10	Measurement of Earth resistance and Earth Leakage Circuit Breaker (ELCB) / Residual Current Circuit Breaker (RCCB).
11	Single phase Uninterrupted power supply connection.
12	Operation and Characteristics of Single-Phase Rectifier.
13	Study and Performance Analysis of a Single-Phase Inverter.

Reference Books:

1	Basic Electrical Engineering- S. K. Sahdev, Pearson Publications, ISBN 978-93-325-4216-7
2	Dr P.S. Bimbhra, Power Electronics - 4th Edition, Kanna Publishers
3	Generation Distribution and Utilization of Electrical Energy by C.L Wadhwa.3rd Edition

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
A2610018	MC	--	--	2	--	40	-	

NSS, SPORTS & YOGA

(Common to AIDS, CIC, CIVIL, CSBS, CSD, ECE, EEE)

Course Objectives:

1.	The objective of introducing this course is to impart discipline, character, fraternity, teamwork, social consciousness among the students and engaging them in selfless service.
2.	To make the students maintain their mental and physical wellness by balancing emotions in their life. It mainly enhances the essential traits required for the development of the personality.

Course Outcomes

S.No	Outcome	Knowledge Level
1.	Understand the importance of discipline, character and service motto.	K2
2.	Explore human relationships by analyzing social problems.	K3
3.	Develop leadership skills and civic responsibilities.	K3
4.	Understand the importance of yoga and sports for physical fitness and sound health	K2
5.	Demonstrate health related fitness components.	K2

SYLLABUS

UNIT-I (10Hrs)	**Orientation** : General Orientation on NSS/NCC/ Scouts & Guides/Community Service activities, career guidance. **Activities:** i Conducting –ice breaking sessions-expectations from the course-knowing personal talents and skills ii Conducting orientations programs for the students –future plans-activities-releasing road map etc. iii Displaying success stories-motivational biopics- award winning movies on societal issues etc. iv Conducting talent show in singing patriotic songs-paintings- any other contribution.
UNIT-II (10 Hrs)	**Nature & Care** **Activities:** i Best out of waste competition. ii Poster and signs making competition to spread environmental awareness. iii Recycling and environmental pollution article writing competition. iv Organising Zero-waste day.

	v Digital Environmental awareness activity via various social media platforms. vi Virtual demonstration of different eco-friendly approaches for sustainable living. vii Write a summary on any book related to environmental issues.
UNIT-III (10 Hrs)	**Community Service** **Activities:** i Conducting One Day Special Camp in a village contacting village-area leaders- Survey in the village, identification of problems- helping them to solve via media-authorities-experts-etc. i Conducting awareness programs on Health-related issues such as General Health, Mental health, Spiritual Health, HIV/AIDS, ii Conducting consumer Awareness. Explaining various legal provisions etc. iii Women Empowerment Programmes- Sexual Abuse, Adolescent Health and Population Education. iv Any other programmes in collaboration with local charities, NGOs etc.
UNIT-IV (10 Hrs)	Yoga Activities: Yoga practices- Asana, Surya Namaskar, Meditation.
UNIT-V (10 Hrs)	Physical Activities: Practicing general and specific warm up, Aerobics. Cardio respiratory fitness, Skipping and Running. Practice one major game and one individual sport(volley ball, basket ball).

Reference Books:	
1.	Nirmalya Kumar Sinha & Surajit Majumder, A Text Book of National Service Scheme Vol; I, Vidya Kutir Publication, 2021 (ISBN 978-81-952368-8-6)
2.	Red Book-National Cadet Corps-Standing Instructions Vol I & II, Directorate General of NCC, Ministry of Defence, New Delhi
3.	Davis M. L. and Cornwell D. A., "Introduction to Environmental Engineering", McGraw Hill, New York 4/e 2008
4.	Masters G. M., Joseph K. and Nagendran R. "Introduction to Environmental Engineering and Science", Pearson Education, New Delhi. 2/e 2007
5.	Ram Ahuja. Social Problems in India, Rawat Publications, New Delhi.
6.	The Sports Rules Book/ Human Kinetics with Thomas Hanlon. -- 3rd ed. Human Kinetics, Inc.2014
e-Resources	
1.	https://books.google.co.in/books/about/Social_Problems_in_India.html?id=jsvJoAEACAAJ&redir_esc=y
2.	https://openlibrary.org/verify_human?next=/books/OL32142863M/Social_problems_in_India%3Futm_source%3Dchatgpt.com

Regulation:R26	I/IV-B.Tech. II -Semester

ELECTRONICS & COMMUNICATION ENGINEERING

COURSE STRUCTURE

(With effect from 2026-27 admitted Batch onwards)

Course Code	Course Name	Category	L	T	P	Cr	C.I.E.	S.E.E.	Total Marks
A2620001	Differential Equations and Vector Calculus	**BS**	3	0	0	3	40	60	100
A2620401	Applied Chemistry	**BS**	3	0	0	3	40	60	100
A2620402	Data Structures and Algorithms using Python	**ES**	0	1	3	2.5	50	50	100
A2620403	Network Analysis	**PC**	3	0	0	3	40	60	100
A2620012	AI Tools and Applications	**ES**	0	1	3	2.5	50	50	100
A2620404	IT Workshop	**ES**	0	0	3	1.5	50	50	100
A2620405	Network Analysis Laboratory	**PC**	0	0	3	1.5	50	50	100
A2620406	Applied Chemistry Laboratory	**BS**	0	0	3	1.5	50	50	100
A2620019	English for Technical Communication and MOODLES Lab	**HS**	0	0	3	1.5	50	50	100
A2620020	Universal Human Values - II	**MC**	2	0	0	0	50	-	50
		TOTAL	**11**	**2**	**18**	**20**	**470**	**480**	**950**

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
A2620001	BS	3	--	--	3	40	60	3 Hrs.

DIFFERENTIAL EQUATIONS AND VECTOR CALCULUS

(Common for all Branches)

Course Objectives: Students are expected to learn

1.	First order ordinary differential equations and some simple geometrical and physical applications.
2.	Methods of solution of linear higher order ordinary differential equations.
3.	Formation and solution of linear partial differential equations.
4.	Concepts of Gradient, divergence, curl.
5.	Vector integral theorems.

Course Outcomes: After completion of the course, the student will be able to

S.No	Outcome	Knowledge Level
1.	**Apply** the knowledge in simple applications such as Newton's law of cooling, orthogonal trajectories and simple electrical circuits.	K3
2.	**Solve** linear ordinary differential equations of second and higher order which occur in various engineering fields.	K3
3.	**Apply** the appropriate methods of solution for partial differential equations that model physical processes.	K3
4.	**Compute** the gradient of a scalar field, curl and divergence of various vector fields to demonstrate their physical behavior.	K3
5.	**Calculate** the work done against a field, circulation and flux using vector calculus.	K3

SYLLABUS

UNIT-I (10Hrs)	**Differential equations of first order and first degree** Exact equations and equations reducible to exact form (Integrating Factor), Linear differential equations, Bernoulli's equations. Orthogonal Trajectories, Applications: Newton's Law of cooling, Law of natural growth and decay, Electrical circuits: RL & RC circuits.
UNIT-II (10 Hrs)	**Linear differential equations of higher order (Constant Coefficients)** Definitions, homogenous and non-homogenous, complimentary function, particular integral, general solution, Method of variation of parameters – Wronskian. Simultaneous linear equations, Applications to L-C-R Circuit problems and Simple Harmonic motion.
UNIT-III (10 Hrs)	**Partial Differential Equations** Introduction and formation of Partial Differential Equations by elimination of arbitrary constants and arbitrary functions, solutions of first order linear equations using Lagrange's method. Homogeneous and Non-Homogeneous Linear Partial differential equations with constant coefficients.
UNIT-IV (10 Hrs)	**Vector Differentiation** Scalar and vector point functions, vector operator Del, Gradient and applications, Directional derivative, Divergence – Solenoidal and Curl – Irrotational Field – Scalar Potential, Vectoridentities.

UNIT-V (10 Hrs)	**Vector Integration** Line Integral – Circulation – Work done by force, surface integral – flux, Green's theorem in the plane (without proof), Stoke's theorem (without proof), volume integral, Divergence theorem (without proof) and their applications.

Textbooks:	
1.	HigherEngineeringMathematics,B.S.Grewal,KhannaPublishers,2017, 44th Edition.
2.	AdvancedEngineeringMathematics,ErwinKreyszig,JohnWiley&Sons,2018,10thEdition.
Reference Books:	
1.	Thomas Calculus, George B. Thomas, Maurice D. Weir and Joel Hass, Pearson Publishers, 2018, 14th Edition.
2.	Advanced Engineering Mathematics, Dennis G. Zill and Warren S. Wright, Jones and Bartlett, 2018.
3.	Advanced Modern Engineering Mathematics, Glyn James, Pearson publishers, 2018, 5th Edition.
4.	Advanced Engineering Mathematics, R. K. Jain and S. R. K. Iyengar, Alpha Science International Ltd., 2021 5th Edition (9th reprint).
5.	Higher Engineering Mathematics, B. V. Ramana, McGraw Hill Education, 2017
6.	Advanced Engineering Mathematics by H. K Dass, S. ChandPublications,2022, Twenty two Edition (Reprint 2022).
7.	Numerical Python: A Practical Techniques Approach for Industry, Robert Johansson, APress, 2015.

e-Resources	
1.	https://mathworld.wolfram.com/First-OrderOrdinaryDifferentialEquation.html
2.	https://en.wikipedia.org/wiki/Differential_equation
3.	https://en.wikipedia.org/wiki/Partial_differential_equation
4.	https://en.wikipedia.org/wiki/Vector_calculus
5.	https://en.wikipedia.org/wiki/Vector_calculus

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
A2620401	BS	3	--	--	3	40	60	100

APPLIED CHEMISRTY

(For ECE)

Course Objectives:

1.	**Develop** an understanding of quantum chemical principles and molecular orbital concepts relevant to the electronic structure and bonding of molecules.
2.	**Develop** knowledge of electrochemical principles and energy-storage and energy-conversion devices, including batteries, fuel cells.
3.	**Apply** chemical principles to understand the preparation, properties, and applications of semiconductor materials, electronic polymers, photoresist materials, and functional polymers used in electronic devices.
4.	**Develop** knowledge of the principles, working mechanisms, and applications of electrochemical sensors, biosensors, colorimetry, UV-Visible spectroscopy, and IR spectroscopy.
5.	**Develop** an understanding of the synthesis, properties, and technological applications of nanomaterials, including carbon nanomaterials, quantum dots, metallic nanoparticles, and spintronic materials.

Course Outcomes: After successful completion of the course, the student will be able to

S.No	Outcome	Knowledge Level
1.	**Apply** quantum chemistry principles to **analyze** electronic structures, molecular orbitals, bonding, and energy levels of molecules.	K3, K4
2.	**Apply** electrochemical principles to **analyze** the operation and characteristics of batteries, fuel cells.	K3, K4
3.	**Apply** materials chemistry principles to **analyze** the properties and applications of semiconductor materials, organic electronic materials, and functional polymers in electronic devices.	K3, K4
4.	**Apply** the principles of sensors, biosensors, colorimetry, UV-Visible, and IR spectroscopy to **analyze** chemical species and interpret their characterization data.	K3, K4
5.	**Apply** nanochemistry principles to **analyze** the synthesis, properties, and applications of carbon nanomaterials, quantum dots, metallic nanoparticles, and spintronic materials in emerging technologies.	K3, K4

SYLLABUS

UNIT-I (10Hrs)	**Quantum Chemistry** • Fundamentals of quantum chemistry: black-body radiation, photoelectric effect, wave-particle duality, de Broglie wave equation, Heisenberg's uncertainty principle. • Fundamentals of quantum mechanics: Schrödinger wave equation (time independent), significance of Ψ and Ψ^2; particle in a one-dimensional box.

	• Molecular orbital theory: Bonding in homo- and hetero-nuclear diatomic molecules; energy-level diagrams of O2, N2, HF and CO; calculation of bond order. • π-molecular orbitals of butadiene and benzene.
UNIT-II (10 Hrs)	**Electrochemistry and Energy storage devices** • Fundamentals of electrochemistry: Electrochemical cells, Electrode potential, Nernst equation, Types of electrodes-ion-selective and glass electrode, Reference electrodes -Calomel electrode, (Construction and working), Electro-chemical series and its uses Conductometric titrations: strong acid vs strong base titrations. • Batteries: Introduction, classification of batteries, characteristics of battery-capacity, power density, cell balancing and cycle life. • Primary batteries: construction, working and applications of zinc-air and Lithium batteries • Secondary batteries: lead-acid, Lithium-ion batteries—construction, charging-discharging reactions and applications. • Fuel cells: Hydrogen-oxygen; Polymer electrolyte fuel cells (PEM), solid-oxide fuel cell (SOFCs) working, applications and limitations.
UNIT-III (10 Hrs)	**Chemistry of Electronic materials (metals and polymers)** • Intrinsic and extrinsic semiconductors, Preparation of pure semiconductor material (Zone refining), Chemistry of compound semiconductors (GaAs, InP, SiC), Advanced semiconductor materials Organic (Introduction, applications,Conduction mechanism of Polyacetylene) and Perovskite semiconductors (Definition,applications), • Photolithography: Basic principle, applications • Polymers: Thermoplastics and Thermosetting Plastics, PTFE, Bakelite, photoresist polymers (definition, applications), structure and applications of PMMA (Poly methyl methacrylate), Polyimide, BCB (Benzocyclobutene), PBO (Polybenzoxazole).
UNIT-IV (10 Hrs)	**Sensors, Biosensors and Principles of Spectroscopic Techniques** • Sensors: Introduction, terminologies- Transducer, Actuators and Sensors, working principle and applications of electrochemical sensor,(Ampherometric sensors, Potentiometric sensors) . • Biosensors: Components of a Biosensor and applications. • Colorimetric Method: Electromagnetic spectrum, Beer-Lambert's law, Principle and instrumentation of colorimeter, application of colorimeter in iron estimation. • Spectroscopy: UV-Visible spectroscopy: principle, electronic transitions, instrumentation, applications.

	• Infrared (IR) spectroscopy: Introduction, principle, mode of vibrations, instrumentation and applications.
UNIT-V (10 Hrs)	**Chemistry of nanomaterials** • Introduction to Nano chemistry: carbon nanotubes (CNTs), graphene; synthesis methods [chemical vapor deposition (CVD, Arc- discharge method), Applications. • Quantum dots: Synthesis (Sol-Gel), properties, and real time applications (biological imaging and sensing). • Metallic nanoparticles: Preparation (Biological process) and applications of gold, silver nanoparticles. • Applications of nanomaterials in nanoelectronics, interconnects, sensors (optical and biosensors), memory devices. • Materials of Spintronics: Magnetic nanoparticles, ferromagnetic materials, topological insulators.(Basic Concept and applications)
Textbooks:	
1.	Jain, P. C., & Jain, M. (2020). Engineering chemistry (17th ed.). Dhanpat Rai Publishing Company.
2.	Agarwal, S. (2026). Engineering chemistry (5th ed.). Cambridge University Press.
Reference Books:	
1.	Streetman, B. G., & Banerjee, S. K. (2015). Solid state electronic devices (7th ed.). Pearson Education
2.	Sze, S. M., & Ng, K. K. (2006). Physics of semiconductor devices (3rd ed.). John Wiley & Sons.
3.	Dara, S. S., & Umare, S. S. (2018). Engineering chemistry (15th ed.). S. Chand Publishing.
e-Resources	
1.	IEEE Electron Devices Society. (n.d.). IEEE EDS resources. https://eds.ieee.org
2.	SEMI. (n.d.). Semiconductor manufacturing standards. https://www.semi.org
3.	IEEE Photonics Society. (n.d.). IEEE Photonics Society resources. https://www.photonicssociety.org
4.	NPTEL. (n.d.). Engineering chemistry courses. https://nptel.ac.in

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
A2620402	ES	--	1	3	2.5	50	50	3 Hrs.

Data Structures and Algorithms using Python

(For ECE)

Course Objectives:

1	Introduce students to fundamental programming concepts using Python and strengthen their computational problem-solving skills.
2	Develop an understanding of fundamental data structures, their representations, operations and applications.
3	Provide hands-on experience in implementing data structures and algorithms using Python.
4	Develop the ability to analyze algorithms using time and space complexity and compare alternative solutions

Course Outcomes: On completion of this laboratory course, the students will be able to

S.No	Outcome	Knowledge Level
1	Explain and implement fundamental programming constructs, functions, recursion, and basic data structures using Python.	K2
2	Explain and apply arrays, linked lists, stacks, and queues to solve computational problems.	K3
3	Implement and analyze trees, heaps, graphs, and hashing techniques using Python	K3
4	Implement searching and sorting algorithms and analyze their time and space complexity	K3
5	Select and apply suitable data structures and algorithms to solve practical engineering problems.	K5

LIST OF EXPERIMENTS

1	**Unit -1** **Introduction to Python Programming:** Features and applications of Python, Python interpreter and execution environment, Variables, constants and data types, Operators and expressions, Input and output statements, Conditional statements, for and while loops, Nested loops, Strings and string operations, Lists, tuples, sets and dictionaries, Functions and parameter passing, Recursion, Modules and packages, Basic exception handling, Introduction to classes and objects, Introduction to built-in data structures. **Lab Experiments** 1. Write a Python program to demonstrate variables, data types and operators. 2. Write a Python program to find the largest and smallest among given numbers. 3. Write a Python program to check prime numbers and generate Fibonacci series. 4. Write a Python program to find factorial using iteration and recursion. 5. Write a Python program to reverse a number and check whether it is a palindrome.

	6. Write a Python program to perform list operations. 7. Write a Python program to perform matrix addition and multiplication. 8. Write a Python program demonstrating strings, tuples, sets and dictionaries. 9. Write a Python program using user-defined functions. 10. Write a Python program demonstrating recursion
2	**UNIT – 2** **ARRAYS, ALGORITHM ANALYSIS AND LINKED LISTS** Introduction to data structures, Need for data structures, Classification of data structures, Linear and non-linear data structures, Abstract Data Types Introduction to algorithms, Time and space complexity, Big-O, Omega and Theta notations, Arrays and their representation, Array traversal, Insertion and deletion, Searching and updating, Two-dimensional arrays, Sparse matrices Introduction to linked lists, Singly linked lists, Linked-list operations, Applications of linked lists **Experiments using Python** 1. Implement array/list traversal, insertion, deletion and updating 2. Implement sparse matrix representation using lists. 3. Implement a singly linked list using a class. 4. Perform insertion and deletion in a singly linked list. 5. Search for an element in a linked list 6. Reverse a singly linked list.
3	**UNIT – 3** **Linked Lists, Stacks And Queues** Linked Lists: Doubly linked lists, Circular linked lists, Insertion and deletion operations, Applications of linked lists Stacks: Stack ADT, Push, pop and peek operations, Stack using arrays, Stack using linked lists, Applications of stacks- Parentheses matching, Infix, prefix and postfix expressions, Infix-to-postfix conversion, Postfix expression evaluation Queues: Queue ADT, Queue operations, Queue using arrays, Queue using linked lists, Circular queues, Applications of queues **Experiments using Python** 1. Implement a doubly linked list using classes. 2. Implement a circular linked list. 3. Implement a stack using a list. 4. Implement a stack using a user-defined class. 5. Implement a queue using a list/class. 6. Implement a circular queue. 7. Check balanced parentheses using a stack. 8. Convert an infix expression to postfix. 9. Evaluate a postfix expression

4	**UNIT – 4** **Trees and Heaps** Introduction to non-linear data structures, Tree terminology, Root, parent, child, leaf, level and height, Binary trees, Binary tree representation, Tree traversals- Preorder, Inorder, Postorder, Binary Search Trees- BST insertion, BST searching, BST deletion, Introduction to heaps, Min heap, Max heap, Heap insertion and deletion, Heapify, Heap sort Applications of trees and heaps **Experiments using Python** 1. Create a binary tree using classes. 2. Implement preorder traversal. 3. Implement inorder traversal. 4. Implement postorder traversal. 5. Implement a Binary Search Tree. 6. Implement insertion and searching in a BST. 7. Implement deletion in a BST. 8. Implement a min heap. 9. Implement a max heap. 10. Implement heap sort.
5	**UNIT – 5** **Graphs, Hashing, Searching And Sorting** Graphs; Introduction to graphs, Graph terminology, Directed and undirected graphs, Weighted and unweighted graphs, Adjacency matrix, Adjacency list, Breadth First Search, Depth First Search, Applications of graphs Hashing: Introduction to hashing, Hash functions, Hash tables, Collision handling, Linear probing, Quadratic probing, Separate chaining Searching: Linear search, Binary search, Comparison of searching algorithms Sorting: Bubble sort, Selection sort, Insertion sort, Merge sort, Quick sort, Comparison of sorting algorithms **Experiments using Python** 1. Represent a graph using an adjacency matrix. 2. Represent a graph using an adjacency list. 3. Implement Breadth First Search. 4. Implement Depth First Search. 5. Implement hashing using linear probing. 6. Implement hashing using separate chaining. 7. Implement linear search. 8. Implement binary search. 9. Implement bubble sort. 10. Implement selection sort.

	11. Implement insertion sort. 12. Implement merge sort. 13. Implement quick sort
Text Books:	
1	Michael T. Goodrich, Roberto Tamassia and Michael H. Goldwasser, *Data Structures and Algorithms in Python*, 1st Edition, Wiley, 2013.
2	Brad Miller and David Ranum, Problem Solving with Algorithms and Data Structures Using Python, Franklin, Beedle & Associates.
Reference Books:	
1	Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest and Clifford Stein, Introduction to Algorithms, 3rd/4th Edition, MIT Press.
2	Wing, J. M. (2006). Computational thinking. Communications of the ACM, 49(3), 33–35. https://doi.org/10.1145/1118178.1118215
Web References:	
1	Python Software Foundation. (n.d.).The Python tutorial.https://docs.python.org/3/tutorial/
2	W3Schools. (n.d.).Python tutorial.https://www.w3schools.com/python/

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
A2620403	PC	3	--	--	3	40	60	3 Hrs.

NETWORK ANALYSIS

(For ECE)

Course Objectives: Students should learn

1.	To introduce basic laws, mesh & nodal analysis techniques for solving electrical circuits
2.	To impart knowledge on applying appropriate theorem for electrical circuit analysis
3.	To explain transient behavior of circuits in time and frequency domains
4.	To teach concepts of resonance
5.	To introduce open circuit, short circuit, transmission, hybrid parameters and their interrelationship.

Course Outcomes: At the end of the course, students will be able to

S.No	Outcome	Knowledge Level
1.	Analyze the DC electric circuits using Kirchhoff's laws, Network reduction Techniques and network theorems.	K4
2.	Analyze Transient behavior of R-L, R-C and R-L-C circuits using classical and Laplace Transform methods.	K4
3.	Analyze AC circuits in steady state using phasor & impedance concepts by applying nodal and mesh analysis for R-L, R-C and R-L-C circuits.	K4
4.	Analyze resonant circuits using bandwidth & quality factor concepts and coupled circuits using dot convention	K4
5.	Determine two-port network parameters for RLC networks.	K4

SYLLABUS

UNIT-I (10 Hrs)	**Introduction:** Types of circuit components, Types of Sources and Source Transformations, Mesh analysis and Nodal analysis, problem solving with resistances including dependent sources. Network Theorems: Thevenin's, Norton's, Superposition, Max Power Transfer- problem solving using only independent sources.
UNIT-II (10 Hrs)	**Transients & Laplace Transform:** Transients: First order differential equations, Definition of time constants, R-L circuit, R-C circuit with DC excitation. Laplace transform: introduction, Laplace transformation, basic theorems, problem solving using Laplace transform, partial fraction expansion.
UNIT-III (10 Hrs)	**Steady State Analysis of A.C Circuits:** Impedance concept, phase angle, series R-L, R-C, R-L C circuits problem solving. Complex impedance and phasor notation for R-L, R-C, R-L-C problem solving using mesh and nodal analysis.
UNIT-IV	**Resonance and Coupled Circuits:** Introduction, Definition of Q, Series resonance,

(10 Hrs)	Bandwidth of series resonance, Parallel resonance. Coupled Circuits: Self-inductance, Mutual inductance, Coefficient of Coupling.
UNIT-V (10 Hrs)	**Two-Port Networks:** Relationship of two port networks, Z-parameters, Y-parameters, Transmission line parameters, h- parameters, Relationships Between parameter Sets, Parallel & series connection of two port networks, cascading of two port networks. Image and iterative impedances.
Textbooks:	
1.	Engineering Circuit Analysis by William H. Hayt, Jack Kemmerly, Jamie Phillips, Steven M. Durbin, 9th Edition 2020.
2.	Fundamentals of Electric Circuits by Charles K. Alexander and Matthew N. O. Sadiku, McGraw-Hill Education.
Reference Books:	
1.	D. Roy Choudhury, Networks and Systems, New Age International Publications, 2013.
2.	A. Sudhakar, Shyammohan S Palli, Network Analysis, McGraw-Hill Education.
e-Resources	
1.	https://nptel.ac.in/courses/108105159

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
A2620012	ES	--	1	3	2.5	50	50	3Hrs

AI Tools and Applications

(Common to AIML, CIC, CSE, ECE)

Course Objectives: To familiarize students with the landscape of AI tools and their underlying working principles.

1	To develop competence in prompt engineering for effective interaction with generative AI systems
2	To enable students to apply AI tools for data analysis, visualization, and interpretation.
3	To train students in using AI-assisted tools for coding, writing, and productivity tasks.
4	To instill awareness of ethical, legal, and societal considerations in AI tool usage.

Course Outcomes:

S. No	Outcome	Knowledge Level
1	Illustrate the categories, capabilities, and limitations of major AI tools (text, image, code, data).	K3
2	Apply the principles of prompt engineering and demonstrate how generative AI models process instructions.	K3
3	Apply AI tools to solve first-year engineering problems in analysis, writing, and coding tasks.	K3
4	Analyze AI-generated outputs for accuracy, bias, and reliability before use.	K4
5	Evaluate the ethical, academic-integrity, and societal implications of AI tool use, justifying appropriate usage choices.	K4

SYLLABUS

1

Introduction to AI and the AI Tools Landscape What is AI, Machine Learning, and Generative AI — basic definitions and how they relate to one another. Core categories of AI tools: conversational/text, code assistants, data-analysis tools, image-generation tools, and writing/productivity tools. How large language models process and generate responses (conceptual, non-mathematical overview). Criteria for selecting the right AI tool for a given task — accuracy needs, cost, data sensitivity, and output format. Setting up accounts and workspaces; understanding free-tier versus paid-tier capabilities and limitations.

Widely Used AI Tools by Application (for classroom coverage):

Category	Example Tools	Typical Application
Conversational/General-Purpose AI	ChatGPT, Claude, Gemini, Perplexity	Writing, research, coding help, file/data analysis, cited web search
AI Coding Assistants	GitHub Copilot, Cursor, Claude Code	In - IDE code completion, AI Native editing, multi-file agentic coding

	Data Analysis Tools	ChatGPT(Advanced Data Analysis), Microsoft Copilotin Excel, Julius AI	Upload a dataset, generate statistics/charts, spreadsheet Formulas via natural language
	Image Generation	Mid journey, DALL·E 3, Stable Diffusion/Flux	Illustrations, concept art, design mockups
	Writing & Productivity	Grammarly, Gamma, Fathom, Notion AI	Grammar/style checking, slide generation, meeting summaries, task automation
	Audio, Video & Presentation (optional)	Eleven Labs, Runway/Sora, Synthesia	Text-to-speech, AI video generation, script-to-video presentations

Teacher Practical Examples (for classroom demonstration):

1. Live comparison of the same general engineering query (e.g., "explain the engineering design process") answered by two different AI chatbots, highlighting differences in response style and accuracy.

2. Walkthrough of an AI tool's interface — settings, conversation history, file upload, and model selection — using a common platform.

Student Exercise Problems (for practice):

1. Create accounts on two AI platforms and compare their responses to a simple engineering question (e.g., "explain the difference between analysis and design").

2. Prepare a one-page comparison chart of five AI tools by category and primary use case.

3. Write a short reflection identifying one task in your daily academic life where an AI tool could help.

2

Prompt Engineering and Generative AI Tools

Anatomy of a good prompt — context, clear instruction, desired format, and constraints.

Zero-shot, few-shot, and chain-of-thought prompting techniques and when to use each.

Role-based and context-setting prompts (e.g., asking the AI to respond as a reviewer, tutor, or domain expert).

Iterative prompt refinement — diagnosing a poor AI response and rewriting the prompt to fix it.

Using generative AI for brainstorming, summarizing, and drafting technical content.

Teacher Practical Examples (for classroom demonstration):

1. Refine a vague prompt (e.g., "explain project scheduling") into a precise one step-by-step, showing output improvement at each stage.

2. Demonstrate few-shot prompting by providing examples to guide the AI toward a specific output format (e.g., a formatted comparison table).

Student Exercise Problems (for practice):

1. Write three versions of a prompt (basic, detailed, role-based) for the same task and compare outputs.

2. Use an AI tool to summarize a one-page technical article and verify the summary against the original.

3. Draft a chain-of-thought prompt to solve a multi-step numerical/logical problem (e.g., a unit-conversion or estimation problem) and check the reasoning.

3	**AI Tools for Data Analysis and Visualization** AI-assisted spreadsheet and dataset analysis using natural-language commands. Natural-language querying of datasets to extract summary statistics and insights. AI tools for generating charts, graphs, and visual summaries from tabular data. Interpreting AI-generated statistical summaries correctly and recognizing their limitations. Identifying and correcting errors or misinterpretations in AI-generated data analysis. **Teacher Practical Examples (for classroom demonstration):** **1.** Upload a small dataset (e.g., class attendance or survey data) to an AI tool and generate a summary-statistics report and chart using natural-language commands. **2.** Show an example where an AI tool misinterprets data (e.g., treats a numeric code as text), and demonstrate how to verify and correct it. **Student Exercise Problems (for practice):** **1.** Use an AI tool to clean and summarize a small sample dataset (e.g., class attendance or marks data). **2.** Generate three different chart types from the same dataset using AI prompts and justify which is most appropriate. **3.** Cross-check one AI-generated statistical claim manually and report any discrepancy found.
4	**AI Tools for Coding, Writing, and Productivity** AI coding assistants — code generation, explanation, and debugging support for simple programs. AI writing assistants for reports, emails, and technical documentation. AI tools for scheduling, note-taking, and meeting/lecture summarization. AI-assisted presentation and slide-outline generation from a topic prompt. Integrating AI tools into an efficient day-to-day academic and engineering workflow. **Teacher Practical Examples (for classroom demonstration):** **1.** Use an AI coding assistant to generate a simple program (e.g., a temperature converter), then intentionally introduce a bug and have the AI debug it. **2.** Generate a presentation outline and speaker notes from a topic prompt using an AI writing/productivity tool. **Student Exercise Problems (for practice):** **1.** Use an AI coding tool to write and debug a simple program for a given problem statement. **2.** Draft a formal email or short report using an AI writing assistant, then manually edit it for accuracy and tone. **3.** Generate a slide outline for a 5-minute seminar topic using an AI tool and refine it.
5	**Ethics, Reliability, and Emerging Trends** Bias and hallucination in AI outputs, and techniques for detecting and fact-checking them. Academic integrity and appropriate citation of AI assistance in coursework and submissions. Data privacy and security considerations, including obligations under India's Digital Personal Data Protection Act, 2023 (DPDP Act), when uploading personal or institutional data. Copyright and intellectual property considerations in AI-generated text and image content. Emerging trends — multimodal AI and AI agents — and their implications for future engineering practice. **Teacher Practical Examples (for classroom demonstration):**

	1. Show a documented case of AI hallucination (a fabricated fact or citation) and demonstrate how to fact-check it. **2.** Walk through an example of properly citing AI assistance in an academic submission per institutional guidelines. **3.** Walk through a worked example of checking whether AI-generated text or image output raises a copyright/IP concern (e.g., close paraphrasing of a source). **Student Exercise Problems (for practice):** **1.** Identify and document a hallucination or bias instance produced by an AI tool on a factual query. **2**. Draft an "AI usage declaration" for a hypothetical assignment, citing tools and extent of use appropriately. **3.** Review the data-handling/privacy policy of one AI tool used in this course and identify one DPDP Act, 2023 compliance consideration relevant to student data.
Text Books:	
1	Marr, B. (2024). Generative AI in practice: 100+ amazing ways generative artificial intelligence is changing business and society. Wiley.
2	Mollick, E. (2024). Co-intelligence: Living and working with AI. Portfolio/Penguin.
Reference Books:	
1	Selwyn, N. (2022). Education and technology: Key issues and debates (3rd ed.). Bloomsbury.
2	Susskind, R., & Susskind, D. (2022). The future of the professions: How technology will transform the work of human experts (Updated ed.). Oxford University Press.
3	Russell, S., & Norvig, P. (2021). Artificial intelligence: A modern approach (4th ed.). Pearson.
Web Resources:	
1	Google. (n.d.). Google AI essentials. Grow with Google. https://grow.google/ai-essentials/
2	OpenAI. (n.d.). OpenAI documentation and guides. https://platform.openai.com/docs
3	Anthropic. (n.d.). Prompt engineering overview. Claude Docs. https://docs.claude.com/en/docs/build-with-claude/prompt- Engineering/overview
4	UNESCO. (2023). Guidance for generative AI in education and research. https://unesdoc.unesco.org/
5	AICTE. (2025). AICTE inputs on Artificial Intelligence — Year of AI initiative. https://www.aicte.gov.in/downloads/initiatives/AICTE%20inputs%20on%20Artificial%20Intelligence%20-compre hensive.pdf
6	Ministry of Electronics and Information Technology, Government of India. (2023). Digital Personal Data Protection Act, 2023. https://www.meity.gov.in/content/digital-personal-data-protection-act-2023

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
A2620404	**ES**	**--**	**--**	**3**	**1.5**	**50**	**50**	**3 Hrs**

IT Workshop

(For ECE)

Course Objectives:

1.	Understand the basic components of a computer, assemble and disassemble a PC, and install an operating system.
2.	Enable students to connect to and navigate the Internet safely, and to understand netiquette and cyber hygiene practices.
3.	Develop proficiency in office automation tools (Word, Excel, and PowerPoint) for academic and professional documentation, data analysis, and presentation.

Course Outcomes: After completion of the course, the student will be able to

S.No	Outcome	Knowledge Level
1.	Identify computer hardware components and assemble, disassemble and set up a working PC with the required operating system.	K3
2.	Apply basic network, browser, search, and cyber security skills to access the Internet, troubleshoot connectivity problems, and practice safe online communication.	K3
3.	Build formatted technical documents such as certificates, abstracts and newsletters using Word	K3
4.	Test for formulas, functions and charts on spreadsheets for data analysis, and design structured interactive presentations.	K4

SYLLABUS

1	**Identifying Components:** Identify the peripherals of a computer and the components inside the CPU cabinet (motherboard, processor, RAM, SMPS, storage drives such as HDD/SSD/NVMe, and graphics card) and their functions. Draw the block diagram of the CPU with the configuration of each component.
2	**Assembling a PC** : Disassemble a PC and reassemble it back to working condition. The lab instructor verifies the work and conducts a viva; refer to the demonstration video provided with the course content.
3	**Operating System Installation:** Individually install Windows and Linux on a personal computer. The lab instructor verifies the installation and conducts a viva.
4	**Network Connectivity :** Connect the system to a LAN/Wi-Fi and access the Internet by configuring TCP/IP settings (IP address, DNS). Demonstrate access to websites and email, and use basic troubleshooting commands (ping, ipconfig/ifconfig, tracert).

5	**Browsers, Search and Netiquette:** Customize a web browser with bookmarks, pop-up blockers and useful extensions (PDF viewer, ad blocker, password manager). Use search engines effectively with filters and Boolean operators, and demonstrate proper netiquette for online communication.
6	**Cyber Hygiene :** Identify common online threats (phishing, malware, unsafe downloads) and configure the system for safe browsing: HTTPS-only browsing, strong passwords with a password manager, two-factor authentication, and keeping the OS and antivirus updated.
7	**Certificate Creation :** Using MS Word , Create a project certificate covering fonts, drop cap, text effects, character spacing, borders and colours, headers and footers, and date/time options.
8	**Project Abstract:** Create a project document on a topic of your choice, and demonstrate the use of formatting styles, tables, bullets and numbering, text direction, cell alignment, footnotes, hyperlinks, symbols, spell check, and track changes.
9	**Newsletter, and Introduction :** Create a newsletter covering table of contents, newspaper columns, images and clip art, the drawing toolbar and WordArt, image formatting, text boxes, paragraph formatting and mail merge.
10	**Spreadsheet Scheduler and GPA:** Using MS Excel, create a scheduler (gridlines, cell formatting, AutoFill, summation) and a GPA sheet using cell referencing, the AVERAGE, STDEV and COUNT functions, charts, and multiple worksheets.
11	**Spreadsheet Data Analysis:** Perform data analysis using LOOKUP/VLOOKUP, splitting cells, freezing panes, grouping/outlining, sorting, Boolean and logical operators, conditional formatting and a basic pivot table.
12	**Interactive Presentation:** Create an interactive presentation covering slide layouts, WordArt, bullets and autoshapes, hyperlinks, inserting images, audio, video, tables and charts; master layouts (slide, template, notes),views and backgrounds and design templates.
Reference Books:	
1	Computer Fundamentals, Anita Goel, Pearson India Education,2017
2	P. K. Sinha and Priti Sinha, Computer Fundamentals, BPB Publications.
3.	Rajaraman V., Fundamentals of Computers, PHI Learning.
4.	Joan Lambert, Microsoft Office 2019/2021 Step by Step, Microsoft Press.
Web References:	
1.	Virtual Labs – Computer Organisation and Hardware: https://www.vlab.co.in/broad-area-computerscience-and-engineering
2.	AWS Free Tier: https://aws.amazon.com/free/

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
A2620405	PC	--	--	3	1.5	50	50	3 Hrs.

NETWORK ANALYSIS LABORATORY

(For ECE)

Course Objectives: Students will learn about

1	To experimentally verify Ohm's law
2	To experimentally verify various theorems of circuit analysis.
3	To measure parameters of iron cored inductor and two port network.
4	To learn phasor concept of R-L-C Circuit.
5	To measure resonance phenomenon for RLC circuits and create locus diagrams of RL, RC series circuits.

Course Outcomes: At the end of the course, students will be able to

S. No	Outcome	Knowledge Level
1	Experimentally verify ohm's law, Kirchhoff's Laws and mesh & nodal analysis using DC circuits.	K3
2	Experimentally verify the various theorems using DC Circuits.	K3
3	Obtain the parameters of choke coil, phasor diagram of RLC circuit and locus diagrams of RL, RC series circuits	K4
4	Determine Z, Y, H and ABCD parameters of two-port networks.	K4
5	Obtain the resonant frequency, bandwidth and quality factor of series RLC resonant circuit.	K3

The following experiments need to be performed using Hardware.

List of Experiments

1	Ohm's Law and obtain the V-I characteristics of Filament Lamp.
2	Verification of KCL and KVL.
3	Mesh and nodal analysis using DC circuit.
4	Thevenin's theorem
5	Norton theorem.
6	Superposition theorem.
7	Maximum power transfer theorem.
8	Parameters of choke coil.
9	Phasor diagram of RLC circuit.
10	Two Port Network Parameters (Z, Y, h and ABCD).
11	Series Resonance
12	Locus diagram of RL and RC series circuits.

Hardware Requirements:	
Regulated Power supplies, Analog/Digital Function Generators, Digital Multimeters, Decade Resistance Boxes/Rheostats, Decade Capacitance Boxes, Ammeters (Analog or Digital), Voltmeters (Analog or Digital), Active & Passive Electronic Components.	
Reference Books:	
1	Network Analysis – ME Van Valkenburg, Prentice Hall of India, revised 3rd Edition, 2019.
2	Engineering Circuit Analysis by William H. Hayt, Jack Kemmerly, Jamie Phillips, Steven M. Durbin, 9th Edition 2020.

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
A2620406	BS	--	--	3	1.5	50	50	100

APPLIED CHEMISTRY LAB

(For ECE)

Course Objectives:

1	To develop practical skills in chemical analysis using volumetric, electrochemical, spectroscopic, and instrumental techniques for the characterization of chemical, water, and electronic-grade samples.
2	To develop hands-on skills in emerging chemistry applications through the synthesis and characterization of nanomaterials, polymeric dispersions, and electrochemical biosensors, and to relate experimental observations to technological applications.

Course Outcomes: After successful completion of the course, the student will be able to

S.No	Outcome	Knowledge Level
1	**Apply** volumetric, conductometric, potentiometric, pH-metric, and redox titration techniques to determine the concentration and properties of acids, bases, electrolytes, and metal ions in chemical and battery electrolyte samples.	K3
2	**Apply** analytical and instrumental techniques, including colorimetry, Winkler's method, and weight-loss measurements, to determine water-quality parameters, metal-ion concentrations, dissolved oxygen, and corrosion rate.	K3
3	**Apply** colorimetric, spectroscopic, electrochemical, and chemical methods to verify Beer–Lambert law, identify functional groups, synthesize polymers and nanomaterials, and investigate electrochemical processes**.**	K3
4	**Apply** chemical synthesis and characterization techniques along with computational/virtual-laboratory methods to prepare nanomaterials and polymeric materials, study hydrogen production by water electrolysis, and determine electronic energy levels using the particle-in-a-box model.	K3

SYLLABUS

1	Determination of the concentration/strength of sulfuric acid in Pb-acid battery electrolyte
2	Conductometric titration of a strong acid against a strong base.
3	Determination of a strong acid against a strong base by using pH meter.
4	Synthesis of nanomaterial by chemical precipitation method
5	Determination of a weak acid against a strong base by using potentiometer.
6	Determination of the corrosion rate of a mild-steel specimen by the weight-loss method.
7	Synthesis of Bakelite.
8	Estimation of Iron in Mohr’s salt by potassium dichromate.

9	To determine the concentration of Fe^{2+}/Fe^{3+} in a given water or leach liquor sample by colorimetry.
10	Determination of Dissolved oxygen in water samples by Winkler's method.
11	Verification of Beer-Lambert's Law by colorimetry.
12	Identify functional groups present in a given organic or mineral-bound sample from its IR spectrum. (virtual lab)
13	Assemble a simple Zn–carbon (or lead-acid) cell and measure its voltage and approximate capacity
14	Calculate and plot allowed energy levels of an electron confined in a one-dimensional box of varying length using the particle-in-a-box model. (virtual Lab)
Reference Books:	
1	Ahluwalia, V. K., & Dhingra, S. (2017). Comprehensive practical organic chemistry. Universities Press
2	Vogel,A.I.(2012), Vogels text book of Quantitative chemical analysis, Pearson Education
3	Virtual Labs, Ministry of Education, GOI, https://vlab.co.in

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
A2620019	HS	--	--	3	1.5	50	50	3Hrs

English for Technical Communication and MOODLE Laboratory

(Common AIDS, AIML, CIC, CE, CSBS, CSD, CSE, CSIT, ECE, EEE, IT.)

Course Objectives:

1	Enable students apply correct phonological patterns for effective communication.
2	Develop students' ability to employ listening strategies to comprehend audio texts.
3	Enable students participate effectively in group discussions, presentations, debates, and interviews in professional contexts.
4	Enhance students' ability to write cohesively and coherently.
5	Familiarize students with Moodle LMS and digital learning tools for effective use of learning resources

Course Outcomes:

Sl. No	Outcome	Knowledge Level
1	Apply appropriate phonological features to communicate effectively in formal and informal contexts.	K3
2	Apply listening strategies to comprehend audio texts, take notes, and narrate events.	K3
3	Demonstrate effective participation in group discussions, presentations, debates, and interviews to communicate in professional contexts.	K3
4	Develop an ability to write clearly on topics related to technology and workplace communication.	K3
5	Demonstrate proficiency in using Moodle LMS and digital learning tools to access learning resources effectively.	K2

SYLLABUS

PART-A

1	Phonetics & Accent Neutralization
2	Introducing Self and Others
3	Brief Talk in Formal/Informal Contexts, JAM/Extempore
4	Listening Comprehension
5	Audio-based Note Taking & Narrating Events
6	Process Description and Paragraph Writing (Technical Lab Context)
7	Group Discussions and Deliberations
8	Micro Presentations (2-5 Mts)
9	Presenting Logical Arguments – Debates
10	Interviews – Formal & Semi-formal

PART-B Moodle Learning Management System: Digital Learning and Academic Skills	
1	Introduction to Moodle LMS • Introduction to Moodle • Student login and profile management • Moodle dashboard and course navigation • Accessing announcements and course materials • Notifications and calendars
2	Learning Through Moodle • Accessing PDFs, videos, web resources, and lecture materials • Downloading and organizing study materials • Moodle lessons and learning activities • Tracking course completion • Managing deadlines and study schedules
3	Assignments, Quizzes and Assessments • Understanding assignment instructions • Creating and uploading assignments • File formats and file naming • Online quizzes and question types • Viewing marks and teacher feedback • Resubmission and revision
4	Digital Communication and Collaboration • Moodle forums and discussions • Academic email etiquette • Online communication and netiquette • Group activities and peer learning • Collaborative learning tools • Digital presentation skills
Lab Manual:	
1	Propel: A Course Book for Technical English Communication. Cengage Publishers, 2026.
Reference Books:	
1	Richards, Jack C., Jonathan Hull, and Susan Proctor. Interchange Level 3 Student's Book with Self-study DVD-ROM. Vol. 3. Cambridge University Press, 2012.
2	Taylor Grant: English Conversation Practice,Tata McGraw Hill Education India,2016.
3	Hewing's, Martin. Cambridge Academic English (B2).CUP, 2012.
4	J.Sethi & P.V.Dhamija. A Course in Phonetics and Spoken English, (2ndEd), Kindle, 2013
5	Raman Meenakshi,Sangeeta-Sharma.TechnicalCommunication.OxfordPress.2018.

Web Resources:	
	1. www.esl-lab.com 2. www.englishmedialab.com 3. www.englishinteractive.net 4. https://www.britishcouncil.in/english/online 5. http://www.letstalkpodcast.com 6. https://www.youtube.com/c/mmmEnglish_Emma/featured 7. https://www.youtube.com/c/ArnelsEverydayEnglish/featured 8. https://www.youtube.com/c/engvidAdam/featured 9. https://www.youtube.com/c/EnglishClass101/featured 10. https://www.youtube.com/c/SpeakEnglishWithTiffani/playlists 11. https://www.youtube.com/channel/UCV1h_cBE0Drdx19qkTM0WNw 12. https://www.youtube.com/user/letstalkaccent/videos 13. https://www.youtube.com/c/EngLanguageClub/featured 14. https://www.youtube.com/channel/UC_OskgZBoS4dAnVUgJVexc 15. https://www.youtube.com/channel/UCNfm92h83W2i2ijc5Xwp_IA

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
A2620020	MC	2	--	--	--	50	--	--

Universal Human Values – II Understanding Harmony and Ethical Human Conduct

(Common to AIDS, CIC, CIVIL, CSBS, CSD, ECE, EEE)

Course Objectives: The objective of this course is to make the student aware of

1.	Essential complementarity between 'Values' and 'Skills' to ensure sustained happiness and prosperity which are the core aspirations of all human beings.
2.	Harmony in the human being, family, society and nature/existence
3.	Holistic perspective towards life, profession and happiness

Course Outcomes: At the end of this course student will be able to

S.No	Outcome	Knowledge Level
1.	**Explain** the role of value education in achieving basic human aspirations.	K2
2.	**Summarize** needs to obtain harmony in self (I).	K2
3.	**Describe** criteria for human-human relationship and harmony in society.	K2
4.	**Explain** four orders of nature and our existence.	K2
5.	**Interpret** significance of harmony in holistic development	K2

SYLLABUS

UNIT-I (06Hrs)	**Introduction to Value Education**: Right Understanding, Relationship and Physical Facility (Holistic Development and the Role of Education); Understanding Value Education-Practice Session PS1 Sharing about Oneself; self-exploration as the Process for Value Education; Continuous Happiness and Prosperity – the Basic Human Aspirations-Practice Session PS2 Exploring Human Consciousness; Happiness and Prosperity – Current Scenario; Method to Fulfill the Basic Human Aspirations-Practice Session PS3 Exploring Natural Acceptance.
UNIT-II (08Hrs)	**Harmony in the Human Being**: Understanding Human being as the Co-existence of the self and the body; Distinguishing between the Needs of the self and the body -Practice Session PS4 Exploring the difference of Needs of self and body; The body as an Instrument of the self; Understanding Harmony in the self-Practice Session PS5 Exploring Sources of Imagination in the self; Harmony of the self with the body; Programme to ensure self-regulation and Health - Practice Session PS6 Exploring Harmony

	of self with the body.
UNIT-III (06 Hrs)	**Harmony in the Family and Society**: Harmony in the Family – the Basic Unit of Human Interaction; 'Trust' – the Foundational Value in Relationship-Practice Session PS7 Exploring the Feeling of Trust; 'Respect' – as the Right Evaluation-Practice Session PS8 Exploring the Feeling of Respect; Other Feelings, Justice in Human-to-Human Relationship; Understanding Harmony in the Society; Vision for the Universal Human Order- Practice Session PS9 Exploring Systems to fulfil Human Goal.
UNIT-IV (06 Hrs)	**Harmony in the Nature/Existence:** Understanding Harmony in the Nature; Interconnectedness, self- regulation and Mutual Fulfilment among the Four Orders of Nature-Practice Session PS10 Exploring the Four Orders of Nature; Realizing Existence as Co-existence at All Levels; The Holistic Perception of Harmony in Existence-Practice Session PS11 Exploring Co-existence in Existence.
UNIT-V (06 Hrs)	**Implications of the Holistic Understanding – a Look at Professional Ethics**: Natural Acceptance of Human Values; Definitiveness of (Ethical) Human Conduct- Practice Session PS12 Exploring Ethical Human Conduct; A Basis for Humanistic Education, Humanistic Constitution and Universal Human Order; Competence in Professional Ethics-Practice Session PS13 Exploring Humanistic Models in Education; Holistic Technologies, Production Systems and Management Models-Typical Case Studies; Strategies for Transition towards Value-based Life and Profession-Practice Session PS14 Exploring Steps of Transition towards Universal Human Order.
Textbooks:	
1.	The Textbook R R Gaur, R Asthana, G P Bagaria, *A Foundation Course in Human Values and Professional Ethics*, 2nd Revised Edition, Excel Books, New Delhi, 2019. ISBN 978-93-87034-47-1
2.	The Teacher's Manual R R Gaur, R Asthana, G P Bagaria,*Teachers' Manual for A Foundation Course in Human Values and Professional Ethics,* 2nd Revised Edition, Excel Books, New Delhi, 2019. ISBN 978-93-87034-53-2
Reference Books:	

1.	*JeevanVidya: EkParichaya*, A Nagaraj, JeevanVidyaPrakashan, Amarkantak, 1999.
2.	Human Values, A.N. Tripathi, New Age Intl. Publishers, New Delhi, 2004.
3.	The Story of Stuff (Book).
4.	The Story of My Experiments with Truth – by Mohandas Karamchand Gandhi
5.	Small is Beautiful – E. F Schumacher.
6.	Slow is Beautiful – Cecile Andrews
7.	Economy of Permanence – J C Kumarappa
8.	Bharat Mein Angreji Raj – PanditSunderlal
9.	Rediscovering India – by Dharampal
10.	*Hind Swaraj or Indian Home Rule* – by Mohandas K. Gandhi
11.	*India Wins Freedom* – Maulana Abdul Kalam Azad
12.	*Vivekananda* – Romain Rolland (English)
13.	*Gandhi* – Romain Rolland (English)
e-Resources	
1.	https://fdp-si.aicte-india.org/UHV-II%20Class%20Notes%20&%20Handouts/UHV%20Handout%201-Introduction%20to%20Value%20Education.pdf
2.	https://fdp-si.aicte-india.org/UHV-II%20Class%20Notes%20&%20Handouts/UHV%20Handout%202-Harmony%20in%20the%20Human%20Being.pdf
3.	https://fdp-si.aicte-india.org/UHV-II%20Class%20Notes%20&%20Handouts/UHV%20Handout%203-Harmony%20in%20the%20Family.pdf
4.	https://fdp-si.aicte-india.org/UHV%201%20Teaching%20Material/D3-S2%20Respect%20July%2023.pdf
5.	https://fdp-si.aicte-india.org/UHV-II%20Class%20Notes%20&%20Handouts/UHV%20Handout%205-Harmony%20in%20the%20Nature%20and%20Existence.pdf
6.	https://fdp-si.aicte-india.org/download/FDPTeachingMaterial/3-days%20FDP-SI%20UHV%20Teaching%20Material/Day%203%20Handouts/UHV%203D%20D3-S2A%20Und%20Nature-Existence.pdf
7.	https://fdp-si.aicte-india.org/UHV%20II%20Teaching%20Material/UHV%20II%20Lecture%2023-25%20Ethics%20v1.pdf
8.	https://www.studocu.com/in/document/kiet-group-of-institutions/universal-human-values/chapter-5-holistic-understanding-of-harmony-on-professional-ethics/62490385
9.	https://onlinecourses.swayam2.ac.in/aic22_ge23/preview